Elijah Olson

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Education

B.S. Computer Science & Engineering | Minor in Philosophy - *University of California, Merced* —————— 05/2026

- **Relevant Coursework:** Deep Learning, Neural Networks, Natural Language Processing, Algorithm Design and Analysis, Data Structures and Design Patterns, Full-stack Web Development, Natural Image Processing, Tech Ethics, Philosophy of the Human Mind, Cognitive Neuroscience, Contemporary Ethics, Metaphysics, Modern Philosophy, Ancient Philosophy, Calculus I-III, Linear Algebra, Linear Analysis, Discrete Mathematics, Statistics

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Experience

Website Manager and Data Technician | *University of California, Merced* —————— 09/2025 - Current

- Completed entire **fullstack** redesign of the CEC website, improving accessibility and user experience to achieve 22% increase in engagement and 45% increase in event attendance.
- Designed and implemented surveys, experience reports, and automated **data pipelines** for all UC Merced-affiliated community programs, creating streamlined feedback and engagement tracking.

Neural Simulation Architect | *SIMBRAIN* —————— 06/2025 - 08/2025

- Transformed theoretical neural circuit models into fully functional implementations in **Kotlin** & **Java**, enabling researchers to simulate diverse neurological systems across Simbrain's platform.
- Enhanced platform accessibility through library improvements, comprehensive documentation, and **UX** refinements, enabling neuroscientists and students of all skill levels to build functional neural models.

Fullstack Engineer Intern | *Patient Safety* —————— 06/2024 - 08/2024

- Architected backend **data pipelines** in **Python** and **SQL**, automating CSV ingestion with validation, creating infrastructure for medical data management across millions of records.
- Collaborated with full-stack engineers on **Django** and **PostgreSQL** interfaces design to optimize seamless backend-frontend integration, and prioritize **UX** by using **Google Sprites** for familiarity of platform.

Research Assistant | *University of California, Merced* —————— 11/2023 - 03/2024

- Developed data processing scripts in **Python** to extract, clean, and organize large sets of EEG data, identifying and isolating relevant neural features to create structured datasets and visualizations with **Pandas** and **Matplotlib**.
- Collaborated with a graduate researcher to translate processed neurological data into actionable insights about sensory processing in the human brain.

Software Engineer Intern | *Tappity Inc.* —————— 04/2020 - 10/2021

- Collaborated with full-stack engineers using **Python, Javascript,** and **React** to design and implement new experiential learning activities for the science education app, leveraging knowledge from prior child care experience, and ultimately boosting user engagement and information retention among K–8 students.

Projects

- **Violet:** AI-powered personalized course creator using **Gemini LLM, Python, React, Javascript,** and **HTML**. Users input subject, timeline, depth, an d learning style; Violet generates customized curricula with curated resources, guided activities, and personalized exercises. 2nd place hackathon winner.
- **Function Visualization:** A design tool made in **Python** and **Pygame** that renders complex fractals (Mandelbrot, Fibonacci, etc.) by mapping functions to visual outputs, exploring the intersection of math and generative art.

Skills

- **Programming & Dev:** C++, C#, Python, Kotlin, JavaScript, HTML, GDScript, Django, React, SQL, Flask, Pandas
- **AI & Neuroscience:** Deep Learning, Machine Learning, Neural Networks, Computational Neuroscience, Neural Simulation, AI Biases, Hidden-Layer Analysis
- **Philosophic:** Tech Ethics, Philosophy of Mind and Cognition, Ethical AI systems, Contemporary and Modern Ethics
- **Procedural:** Use of AI, Project Engineering, UX, Data Visualization, Technical Documentation, Cross-Functional Communications, Systems Thinking, Human-Centered Design